Topic P2: Forces

P2.1 Motion

Summary

Having looked at the nature of matter which makes up objects, we move on to consider the effects of forces. The interaction between objects leads to actions which can be seen by the observer, these actions are caused by forces between the objects in question. Some of the interactions involve contact between the objects, others involve no contact. We will also consider the importance of the direction in which forces act to allow understanding of the importance of vector quantities when trying to predict the action.

Underlying knowledge and understanding

From their work in Key Stage 3 Science, learners will have a basic knowledge of the mathematical relationship between speed, distance and time. They should also be able to represent this information in a distance-time graph and have an understanding of relative motion of objects.

Common misconceptions

Learners can find the concept of action at a distance challenging. They have a tendency to believe that a velocity must have a positive value and have difficulty in associating a reverse in direction with a change in sign. It is therefore importa to make sure learners are knowledgeable about the vector–scalar distinction.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM2.1i	recall and apply: distance travelled (m) = speed (m/s) × time (s)	M1a, M2b, M3a, M3b, M3c, M3d, M4a, M4b, M4c, M4d, M4e
PM2.1ii	recall and apply: $\text{acceleration (m/s}^2\text{)} = \dfrac{\text{change in velocity (m/s)}}{\text{time (s)}}$	M1a, M3a, M3b, M3c, M3d
PM2.1iii	apply: $(\text{final velocity (m/s)})^2 - (\text{initial velocity (m/s)})^2 = 2 \times \text{acceleration (m/s}^2\text{)} \times \text{distance (m)}$	M1a, M3a, M3b, M3c, M3d
PM2.1iv	recall and apply: $\text{kinetic energy (J)} = \frac{1}{2} \times \text{mass (kg)} \times (\text{speed (m/s)})^2$	M1a, M3a, M3b, M3c, M3d

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
P2.1a	describe how to measure distance and time in a range of scenarios				
P2.1b	describe how to measure distance and time and use these to calculate speed	from graphs	M4a, M4b, M4c, M4d, M4f	WS1.2b, WS1.2e, WS1.3a, WS1.3b, WS1.3c, WS1.3g, WS1.3h, WS1.3i, WS2a, WS2b, WS2c, WS2d	Calculations of the speeds of learners when they walk and run a measured distance. Investigation of trolleys on ramps at an angle and whether this affects speed. (PAG P3)
P2.1c	make calculations using ratios and proportional reasoning to convert units and to compute rates	conversion from non-SI to SI units	M1c, M3c		
P2.1d	explain the vector–scalar distinction as it applies to displacement and distance, velocity and speed				
P2.1e	relate changes and differences in motion to appropriate distance-time, and velocity-time graphs; interpret lines and slopes		M4a, M4b, M4c, M4d	WS1.3a	Learners to draw displacement-time and velocity-time graphs of their journey to school. (PAG P3)
P2.1f	**interpret enclosed area in velocity-time graphs**		M4a, M4b, M4c, M4d, M4f		

P2.1g	calculate average speed for non-uniform motion		M1a, M1c, M2b, M3c		
P2.1h	apply formulae relating distance, time and speed, for uniform motion, and for motion with uniform acceleration		M1a, M1c, M2b, M3c	WS1.2b, WS1.2e, WS1.3a, WS1.3b, WS1.3c, WS1.3g, WS1.3h, WS1.3i, WS2a, WS2b, WS2c, WS2d	Investigation of acceleration. (PAG P3)